L1 SOC ALERT TRIAGE REPORT

Investigation of Linux Privileged Group Membership Change

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1. OBJECTIVE & EXECUTIVE SUMMARY

As part of my Week 5 practical SOC analyst training at DG Interns Hub, I investigated an administrative alert regarding a user account modification on our Ubuntu server (`ubuntu-soc`). Modifying system groups or adding user accounts to privileged security groups can be a technique used by adversaries to establish persistence or elevate access.

> **Investigation Summary & Verdict**
>
> Log analysis in Splunk Enterprise identified **4 related security log entries** generated by the command `usermod -aG adm splunk` executed by user `gunjan` via `sudo` on 5 November 2026 at 19:09 UTC.
>
> **Finding:** This activity was an *authorized, benign administrative action* performed intentionally during my lab setup to allow the `splunk` service account to read `/var/log/auth.log`. No unauthorized privilege escalation or malicious activity took place.

2. TOOLS & ENVIRONMENT DETAILS

Component	Configuration Parameters
SIEM Platform	Splunk Enterprise 10.6.1 (Standalone Deployment)
Operating System	Ubuntu Linux (Oracle VirtualBox VM)
Target Hostname	`ubuntu-soc`
Log File Source	`/var/log/auth.log`
Splunk Index & Type	Index: `main` | Sourcetype: `linux_secure`
Executing User	`gunjan` (UID: 1000)
Modified Account	`splunk` (Service Account)

3. PRACTICAL WORK & EVIDENCE COLLECTION

Step 1: Filtering User Modification Activity

I started my investigation in Splunk Web by searching for account modification commands using the keyword `"usermod"` in the Linux security log index:

```
index=main sourcetype=linux_secure "usermod"
```

This search returned exactly **4 events** logged within the selected timeframe.

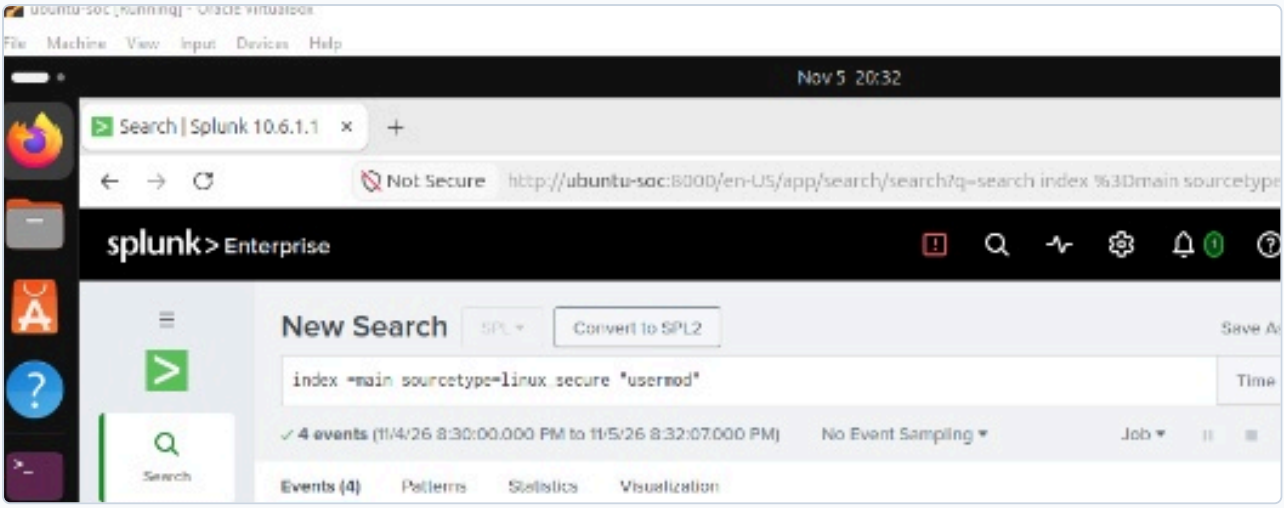


Figure 1: Initial Splunk search returning 4 matching `usermod` events from `/var/log/auth.log`.

Shows 4 raw events captured under `index=main sourcetype=linux_secure "usermod"` on host `ubuntu-soc`.

Step 2: Table Formatting and Timeline Extraction

To perform detailed timeline analysis and clearly examine the command string, host, and raw log messages, I executed a structured table query:

```
index=main sourcetype=linux_secure "usermod" | table _time host _raw | sort -_time
```

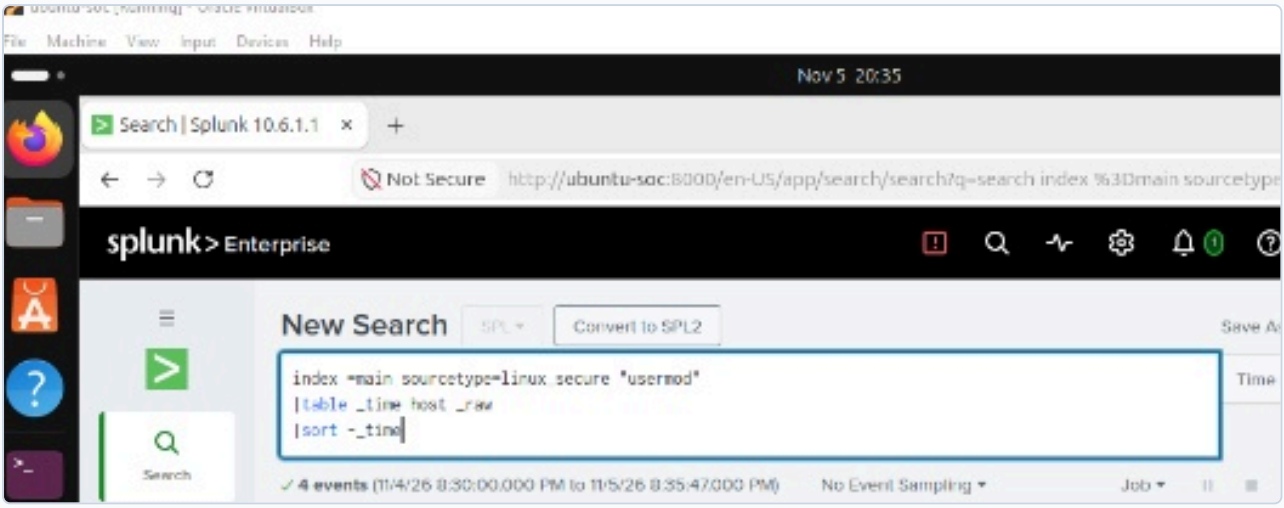


Figure 2: Formatted table view detailing event timestamps, host, and raw log payloads.

Extracted tabular output displaying the exact sequence of commands and system responses during group modification.

Chronological Event Timeline:

MITRE ATT&CK Mapping Evaluation:

If this command had been executed by an unauthorized entity or external compromised account, it would map to:

> **Tactic:** Persistence (TA0003) / Privilege Escalation (TA0004) **Technique: T1098.007 – Account Manipulation: Additional Local or Domain Groups**

> **Assessment Note: No Adversary Technique Established**
> Because this action was confirmed as a legitimate, local administrative configuration step performed during SIEM setup, **no malicious MITRE ATT&CK technique was executed in this instance**.

Timestamp (+05:30)	Process / Utility	Raw Event Log Details (`_raw`)
19:09:47.599	sudo	sudo: gunjan : TTY=pts/0 ; PWD=/home/gunjan ; USER=root ; COMMAND=/usr/sbin/usermod -aG adm splunk